

# Liquidity and market efficiency<sup>☆</sup>

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## Abstract

Short-horizon return predictability from order flows is an inverse indicator of market efficiency. We find that such predictability is diminished when bid-ask spreads are narrower, and has declined over time with the minimum tick size. Variance ratio tests suggest that prices were closer to random walk benchmarks in the more liquid decimal regime than in other ones. These findings indicate that liquidity stimulates arbitrage activity, which, in turn, enhances market efficiency. Further, as the tick size decreased, open-close/close-open return variance ratios increased, while return autocorrelations decreased. This suggests an increased incorporation of private information into prices during more liquid regimes.

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## 1. Introduction

As recognized at least since Hillmer and Yu (1979), Epps (1979), and Patell and Wolfson (1984), evidence that a market is quite efficient (Fama, 1970) over a daily horizon does not preclude inefficiencies at shorter horizons. This is because investors need time to absorb and act on new information. Indeed, recent papers, e.g., Cushing and Madhavan (2000), and Chordia, Roll, and Subrahmanyam (2005) document that short-horizon returns are predictable from past order flows. The determinants of this short-horizon predictability deserve a thorough investigation by finance scholars.

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Short-horizon return predictability should be diminished by arbitrage trading, which should be more extensive and effective during times in which the market is more liquid. Recent evidence indicates that liquidity does vary over time and that market conditions can occasionally be so turbulent that effective trading costs may jump dramatically and liquidity can decline or even disappear.<sup>1</sup> An important but hitherto unaddressed empirical question is whether such fluctuations in liquidity are related to variations in the degree of market efficiency.<sup>2</sup> Further, successive reductions in the minimum tick size have led to exogenous decreases in bid-ask spreads (see, for example, Bessembinder, 2003; or Chordia, Sarkar, and Subrahmanyam, 2005), and a related issue is whether such secular increases in liquidity have enhanced market efficiency. In this paper we address these questions by using return, order flow, and liquidity data for a large sample of NYSE stocks over a ten-year period (1993–2002).

Other studies explore the return/order flow relation, but around specific events or over short periods of time. Sias (1997) analyzes order imbalances in the context of institutional buying and selling of closed-end funds; Lauterbach and Ben-Zion (1993) and Blume, MacKinlay, and Terker (1989) study order imbalances around the 1987 crash; and Lee (1992) does the same around earnings announcements. Chan and Fong (2000) investigate the impact of order imbalances on the relation between stock volatility and volume, using data for about six months. Hasbrouck and Seppi (2001) and Brown, Walsh, and Yuen (1997) study order imbalances for 30 and 20 stocks, over one and two years, respectively. In comparison, we study return predictability from order flows in a comprehensive sample of actively traded stocks over more than 2,000 trading days.

A separate line of research explores a link between liquidity and stock returns by way of a premium demanded by investors for trading in illiquid stocks. For example, Amihud and Mendelson (1986) and Jacoby, Fowler, and Gottesman (2000) provide theoretical arguments and empirical evidence to support the existence of liquidity premia in the cross-section. In addition, Jones (2001) and Amihud (2002) show that liquidity predicts expected returns in the time-series, and Pastor and Stambaugh (2003) and Acharya and Pedersen (2005) find that expected stock returns are cross-sectionally related to liquidity risk. In our paper, we explore a distinctly different link between liquidity and asset prices by asking whether liquidity is associated with an enhanced degree of intraday market efficiency.

Our investigation can be best cast in terms of competing hypotheses on how liquidity and market efficiency may be related. Note first that illiquidity does not necessarily imply any return predictability from order flows. In Glosten and Milgrom (1985) or Kyle (1985), for instance, markets are illiquid because prices for buying and selling shares are different, but prices also are martingales because market makers are risk-neutral. Thus, in these models, current prices and order flows do not predict future returns. Departing from these paradigms, there are at least two ways in which return predictability from order flows can arise. First, if market makers have limited risk-bearing capacity, persistent asymmetric orders will lead to predictable returns due to temporary deviations of prices from fundamental values. Agents such as floor brokers and floor traders who are able to detect such deviations may submit arbitrage trades, which may speed up the convergence of prices to fundamental values. However, the proclivity of these traders to submit arbitrage orders would be negatively influenced by illiquidity. This mechanism creates a link between liquidity and efficiency. Alternatively, risk-neutral but cognitively challenged market makers might misreact to the information content of the order flow (Barberis, Shleifer, and Vishny, 1998) and thereby fail to eliminate return predictability. The mispricing would create an incentive for outside agents to gather information about order flow. If they do so and trade on such information, market makers will face increased adverse selection, and as a result the market may be less liquid. At the same time, prices may be more efficient because more information is impounded into the stock price. This argument suggests that increased efficiency may be associated with less liquidity. Our analysis may then be viewed as a test of these competing hypotheses.

For stocks that traded every day on the NYSE from 1993 to 2002, our empirical results include the following: (a) Intraday market efficiency is intimately linked to daily liquidity; specifically, the forecastability of returns from lagged order flows is significantly diminished on days when the market is more liquid;

<sup>1</sup>“One after another, LTCM’s partners, calling in from Tokyo and London, reported that their markets had dried up. There were no buyers, no sellers. It was all but impossible to maneuver out of large trading bets.”—*Wall Street Journal*, November 16, 1998.

<sup>2</sup>Huang and Stoll (1994) show that intraday returns are predictable from past returns, but do not relate this phenomenon to liquidity. Boehmer, Kelley, and Pirinsky (2005) consider the impact of institutional investors on intraday price discovery.

(b) efficiency has improved with the reduction in the minimum tick size, which corresponds to a decrease in bid-ask spreads; (c) variance ratio tests using intraday and daily midquote returns indicate that prices were closer to a random walk benchmark in the more liquid decimal regimes than in the regimes with higher minimum tick sizes.

The notion of efficient markets (Fama, 1970) emphasizes a lack of return predictability as the criterion for efficiency, while the microstructure literature emphasizes a distinctly separate measure of financial market quality, namely, the amount of private information reflected in prices. As Kyle (1985) points out, even semi-strong efficient prices can reflect varying degrees of private information. Intuitively, markets that become more liquid due to secular events, such as a reduction in tick size, may incorporate private information more readily because an exogenous decrease in trading costs may stimulate trading on information about fundamentals (Admati and Pfleiderer, 1988). We provide evidence of this phenomenon by reporting changes across three tick size regimes in two information-associated statistics, (a) open-to-close versus close-to-open return variances and (b) first-order daily return autocorrelations. The idea is to check whether improved liquidity associated with an exogenous reduction in the tick size enhances informational efficiency by encouraging trading on private information.

We find that open-to-close/close-to-open variance ratios increased and first-order autocorrelations decreased (strongly for smaller firms) as the tick size fell. As French and Roll (1986) suggest, an increase in these variance ratios could be caused either by more mispricing or by increased informational efficiency. Increased mispricing after a tick size reduction should be associated with higher serial correlation of returns. However, return autocorrelations decrease with the tick size. Thus, the evidence points to an increase in private information trading in the lower tick size regimes.

Overall, the data suggest that the liquidity improvement following the tick size reductions has not only accompanied an increase in the efficiency of accommodating order flows, but has also enhanced informational efficiency by allowing better incorporation of private information into prices.

The rest of the paper is organized as follows. Section 2 develops our economic arguments and describes the data. Section 3 presents the intraday evidence. Section 4 presents the variance ratio analysis and Section 5 concludes.

## 2. Economic reasoning and data description

### 2.1. Motivation

There are at least two competing reasons for why market efficiency may be related to liquidity. First, suppose that market makers have limited risk-bearing capacity and/or face inventory financing constraints. In this situation, price pressures induced by incoming order flows may cause midquotes to temporarily diverge from fundamental (“full-information”) values. This implies that returns may be predictable from past order flows over short horizons.<sup>3</sup> Agents who continuously monitor the market, such as day traders, floor traders, or floor brokers trading for their own account, may detect the divergence of midquotes from true asset values. They may then submit arbitrage orders to profit from the temporary deviation of midquotes from their full-information counterparts.<sup>4</sup>

Note that if the orders of arbitrageurs arrive in sufficient quantities and in a timely manner, they would tend to quickly reduce market makers’ excess inventories and thus cause midquotes to adjust speedily to initial imbalance shocks. This adjustment would tend to diminish return predictability, as argued by Chordia, Roll, and Subrahmanyam (2005). However, the inclination of arbitrageurs to submit such orders would be greater when the bid-ask spread, a measure of illiquidity, is low.<sup>5</sup> This argument suggests that more liquid markets should exhibit less pronounced return predictability and vice versa.

<sup>3</sup>See Stoll (1978). Chordia and Subrahmanyam (2004) show that price pressures due to positively autocorrelated imbalances induce a positive relation between returns and lagged imbalances.

<sup>4</sup>These orders would likely be either marketable limit or market orders to assure speedy execution since the deviation of quote midpoints from full information values would be short-lived in actively traded stocks.

<sup>5</sup>See Peterson and Sirri (2002) for detailed documentation of the effective spreads paid by market orders as well as marketable limit orders, and for evidence that trades are sensitive to such costs of transacting. Brennan and Subrahmanyam (1998) also show that trading strategies are affected by illiquidity.

Although the above argument suggests that liquidity reduces the predictability of returns from order flow, there are competing hypotheses. Suppose that market makers underreact to order flow because of cognitive limitations (Barberis, Shleifer, and Vishny, 1998). In this situation, other agents may find it profitable to collect information about order flow and trade on it, thereby increasing efficiency by causing prices to react more fully to the order flow. The presence of such agents, however, would increase the adverse selection problem of the market maker, which means that market liquidity could decrease. Thus, in this scenario, illiquidity would be accompanied by greater market efficiency. And there is yet a third hypothesis, that if outsider assistance is not needed and rational (designated and de facto) market makers promptly absorb imbalances by updating quotes, then illiquidity may bear *no* relation with return predictability from order flows. Our empirical work can be viewed as resolving the tension among these three arguments.

## 2.2. Imbalance and return data

Since the serial dependence in returns is close to zero for active stocks over a daily horizon,<sup>6</sup> an investigation of the relation between liquidity and the efficiency-creating process must focus on intraday trading. An initial research choice involves the length of the intraday interval over which returns and order flows are measured. We decide to focus on five-minute intervals for the following two reasons. First, though shorter intervals are technically possible, nontrading becomes an issue. Second, though longer intervals are feasible, short-lived market inefficiencies would probably be less conspicuous over such intervals, since a predictive relation between order imbalances and future returns is unlikely to last very long. Five minutes seems like a good compromise.<sup>7</sup> Also, we exclude small stocks from our study because of the inherent difficulty in measuring serial dependence at the five-minute horizon when trading is infrequent.

The sample is selected as follows. At the beginning of each year from 1993–2002, all NYSE firms listed on CRSP are sorted on market capitalization, and the largest 500 are noted. Of these, 193 traded every day on the NYSE over the entire period and are retained in the sample. The 1993–2002 period is selected because (a) transactions data are available from the TAQ (Trade and Automated Quotations) database recorded by the Exchange for this period, and (b) this period spans significant changes in the minimum tick size, which was reduced from \$1/8 to \$1/16 during 1997 and was reduced further to one cent by January 2001.<sup>8</sup> The tick size changes are exogenous and provide a natural experiment to test the impact of liquidity on return predictability. Thus, we hope to discern changes in the price formation process during the years preceding and following these events. Future investigations should extend the analysis to smaller firms and other years, exchanges, and countries.

To calculate order imbalances, we apply the Lee and Ready (1991) algorithm, which requires that each transaction be matched to a bid-ask quote.<sup>9</sup> We first filter the trade and quote transactions data as in Chordia, Roll and Subrahmanyam (2001). Then, following Lee and Ready (1991), from 1993 to 1998 inclusive, we set the matching quote to be the first quote at least five seconds prior to the trade. Due to a generally accepted decline in reporting errors in recent years (see, for example, Madhavan, Richardson, and Roomans, 2002), after 1998 the matching quote is simply the first quote prior to the trades. We apply the Lee and Ready (1991) method to the matched trade-quote sample and thereby obtain an estimate of whether a particular trade was buyer- or seller-initiated. Order imbalance for each stock over any time interval can then be calculated variously as the number of buyer- less the number of seller-initiated trades divided by the total number of trades (*OIB#*), or the dollars paid by buyer-initiators less the dollars received by seller-initiators divided by the

<sup>6</sup>See, for example, Chordia, Roll, and Subrahmanyam (2005, p. 273).

<sup>7</sup>Andersen, Bollerslev, Diebold, and Ebens (2001) also use a five-minute interval when computing daily variances from high frequency data.

<sup>8</sup>Ball and Chordia (2001) show that for the largest stocks, more than half the bid-ask spread in 1996 was due to the impact of rounding onto the tick grid. Also, Chordia and Subrahmanyam (1995) argue that a reduction in the tick size would result in more competition and less payment for order flow, thus causing orders to flow to the least-cost providers of market making services.

<sup>9</sup>The Lee–Ready algorithm classifies a trade as buyer- (seller-) initiated if it is closer to the ask (bid) of the prevailing quote. If the trade is exactly at the midpoint of the quote, a “tick test” is used whereby the trade is classified as buyer- (seller-) initiated if the last price change prior to the trade is positive (negative). Note that a limit order is most often the passive side of the trade; i.e., the noninitiator.

total dollars traded (*OIB\$*).<sup>10</sup> The first of these order imbalance measures disregards the size of the trade, counting small orders equally with large orders. The second measure weights large orders more heavily. Our results are similar using either measure; therefore, for the most part, we focus on *OIB\$*.

Order imbalances are computed over all trades within each time interval. For example, contemporaneous *OIB\$* during the five minutes ending at 9:50 a.m. contains the dollar value of buyer-initiated trades less the dollar value of seller-initiated trades between 9:45:01 a.m. and 9:50:00 a.m. The corresponding lagged five-minute *OIB\$* involves the accumulation between 9:40:01 a.m. and 9:45:00 a.m.

We now turn to the calculation of returns over the short (five-minute) intervals that are the focus of our study. In computing such returns, several methods can potentially be used. One alternative is to use transaction prices to compute returns, but such returns are affected by bid-ask bounce. Another alternative is to compute returns from midquotes that are matched to the last transaction during a particular interval. However, because of heterogeneities in trading frequencies across stocks, this procedure would have led to returns being computed over intervals of different lengths for different stocks, confounding the interpretation of the results. Keeping these issues in mind, we present results from returns that are computed by using the midpoints of the quotes prevailing at the end of each five-minute interval. When calculations involve lagged values, the first five-minute interval of each trading day is discarded because it would have been related to a lagged interval from the previous trading day.

### 2.3. Liquidity data

Measures of aggregate illiquidity are constructed from individual firm bid/ask spreads. The effective spread is simply twice the absolute difference between the transaction price and the prevailing quote midpoint, and is obtained from the trade-quote matched sample described in the previous subsection. In a robustness check reported later, we also use the quoted spread, which is computed as the difference between the transaction-matched ask and bid quotes. We compute the market-wide illiquidity indicators as follows. First, for each stock, we calculate the mean effective and quoted spreads over each trading day.<sup>11</sup> These are then value-weighted and averaged across stocks (with market capitalization at the end of the previous year used to calculate weights) to obtain the aggregate measures.

The sample spans three distinct liquidity subperiods: (i) January 4, 1993–June 23, 1997; (ii) June 24, 1997–January 28, 2001; and (iii) January 29, 2001–December 31, 2002. These correspond to the eighth, sixteenth, and decimal tick size regimes, respectively. Table 1 presents summary statistics associated with the illiquidity, imbalance, and return measures separately for each of the tick size regimes.

We find that illiquidity has declined sharply across the three periods; thus, the average effective spread was 12 cents in the eighths regime, eight cents in the sixteenths regime, and three cents in the decimal regime. The order imbalance in number of transactions is fairly stable across the three regimes. While the average dollar imbalance is higher in the decimal period than in the other two regimes,<sup>12</sup> average returns in the post-decimal period were mediocre relative to the prior period. This result is intriguing because, based on earlier research that documents a strong imbalance-return relation (e.g., Chordia, Roll, and Subrahmanyam, 2002), we would expect patterns in order flows to mimic patterns in returns. However, the increased imbalance in the decimal period may be due to individual investors who trend-chase by extrapolating returns from the prior bull market (Griffin, Nardari, and Stulz, 2007). These trades would be uninformative and consequently be unrelated to long-horizon returns.

<sup>10</sup>The scaling of imbalance allows for the notion that large firms would have large dollar imbalances simply because they would have larger volume. However, this raises the concern that the results relating imbalance to returns may be influenced by a relation between returns and total volume (see, e.g., Gervais, Kaniel, and Mingelgrin, 2001), which appears in the denominator of our measure. We also experimented with an alternative scaling method, which involves scaling the dollar imbalance by the average trading volume across five-minute periods throughout the relevant time period. Results from this alternative scaling method are substantively similar to those presented here.

<sup>11</sup>Note that the alternative liquidity proxies of Amihud (2002) and Pastor and Stambaugh (2003) are not readily computable as daily measures.

<sup>12</sup>The sample sizes are equal to the number of five-minute trading intervals during each period, excluding missing observations for all stocks (e.g., the market was closed after the terrorist attack on September 11, 2001) or, occasionally, when one of the three size groups (to be reported upon later) is missing a five-minute interval; approximately 0.06% of the total number of five-minute intervals is associated with missing data. The retained intervals are identical in every calculation.



Table 1

## Summary statistics

Quoted and effective spreads (*QSPR* and *ESPR*, respectively) and order imbalances (*OIB*) are from the NYSE TAQ database. Spreads are averaged across the day for each stock and then value weighted across stocks using market capitalizations at the beginning of each calendar year. *OIB#* is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during a five-minute trading interval. *OIB\$* is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during a five-minute trading interval. *N* is the number of trading days and *n* is the number of five-minute trading intervals. The sample spans the years 1993–2002 inclusive, and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993–June 23, 1997, the sixteenths June 24, 1997–January 28, 2001, and the decimal regime spans the remainder of the sample period.

|                    |                    | QSPR(\$) | ESPR(\$) | OIB#   | OIB\$  |
|--------------------|--------------------|----------|----------|--------|--------|
| Entire sample      | Mean               | 0.1350   | 0.0896   | 0.0594 | 0.0622 |
| <i>N</i> = 2519    | Median             | 0.1447   | 0.0916   | 0.0613 | 0.0705 |
| <i>n</i> = 195,447 | Standard Deviation | 0.0464   | 0.0314   | 0.0925 | 0.251  |
| Eighths regime     | Mean               | 0.1742   | 0.1176   | 0.0568 | 0.0458 |
| <i>N</i> = 1,130   | Median             | 0.1710   | 0.1175   | 0.0576 | 0.0520 |
| <i>n</i> = 87,822  | Standard Deviation | 0.0093   | 0.0031   | 0.166  | 0.293  |
| Sixteenths regime  | Mean               | 0.1305   | 0.0838   | 0.0598 | 0.0705 |
| <i>N</i> = 908     | Median             | 0.1275   | 0.0842   | 0.0618 | 0.0763 |
| <i>n</i> = 70,446  | Standard Deviation | 0.0130   | 0.0069   | 0.139  | 0.219  |
| Decimal regime     | Mean               | 0.0515   | 0.0349   | 0.0658 | 0.0852 |
| <i>N</i> = 481     | Median             | 0.0489   | 0.0333   | 0.0663 | 0.0880 |
| <i>n</i> = 37,179  | Standard Deviation | 0.0114   | 0.0073   | 0.114  | 0.192  |

#### 2.4. Construction of five-minute return/imbalance portfolios

Recall that we use the midpoint of the bid-ask quotes prevailing at the end of the time interval to construct our returns. By using quotes that are not matched to specific trades, we mitigate the problem of nonsynchronous trading. However, if the quotes are updated too infrequently (for example, if they tend to be updated properly only when a trade occurs), the effect of nonsynchronous trading may not be fully removed. In this situation, stocks trading with a lag could induce a spurious apparent forecasting relation between *portfolio* imbalances and future *portfolio* returns. Although we could potentially use individual security data, which would not be plagued by nonsynchronicity, these data are excessively noisy.<sup>13</sup> To alleviate problems with noisy individual stock data, we conduct the analysis using value-weighted portfolios (with market capitalizations at the end of the previous year used to calculate weights). In addition, to address any residual problems arising from nonsynchronous trading, we exclude stocks that did not trade at time  $t-1$  from the portfolio constructed at time  $t$ .

Table 2 presents correlations among five-minute returns and order imbalances. Returns exhibit high contemporaneous correlations with both imbalance measures, and the correlations are reasonably stable across the three liquidity subperiods.

### 3. The intraday evidence

#### 3.1. Basic regressions

Table 3 reports a simple regression of five-minute returns on past five-minute order imbalances measured by the number of transactions (*OIB#*) and by dollars (*OIB\$*). Consistent with Chordia, Roll, and Subrahmanyam (2005), lagged *OIB* is a significant predictor of five-minute returns. This result holds for both measures of imbalance, though the coefficient is larger for *OIB#*. A one-standard deviation move in *OIB\$* increases

<sup>13</sup>In any case, we consider results from individual stock-level analyses in Section 3.3.

Table 2

Contemporaneous correlations for five-minute trading intervals

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. Five-minute returns are computed using the midpoints of the first and last quotes within each five-minute trading interval; *n* is the number of five-minute intervals. *OIB#* is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during a five-minute interval. *OIB\$* is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during a five-minute interval. The sample spans the years 1993 to 2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993 to June 23, 1997, the sixteenths June 24, 1997 to January 28, 2001, and the decimal regime spans the remainder of the sample period.

|                    |       | Return | OIB#  |
|--------------------|-------|--------|-------|
| Entire Sample      | OIB#  | 0.534  |       |
| <i>n</i> = 195,447 | OIB\$ | 0.504  | 0.681 |
| Eighths Regime     | OIB#  | 0.503  |       |
| <i>n</i> = 87,822  | OIB\$ | 0.498  | 0.666 |
| Sixteenths Regime  | OIB#  | 0.650  |       |
| <i>n</i> = 70,446  | OIB\$ | 0.618  | 0.731 |
| Decimal Regime     | OIB#  | 0.595  |       |
| <i>n</i> = 37,179  | OIB\$ | 0.575  | 0.630 |

Table 3

Predictive regressions of five-minute returns on lagged order imbalance, 1993–2002

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. Five-minute returns are computed using the mid-points of the first and last quotes within each five-minute trading interval. The return in interval *t* is denoted *Return<sub>t</sub>*. The variable *OIB#<sub>t</sub>* is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during five-minute interval *t*. *OIB\$<sub>t</sub>* is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during five-minute interval *t*. The sample spans the 1993 through 2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The first five-minute interval of each day is used only in the lagged regressors, and the sample size is 192,928.

| Dependent Variable: <i>Return<sub>t</sub></i> |             |        |                     |
|-----------------------------------------------|-------------|--------|---------------------|
|                                               | Coefficient |        | <i>t</i> -Statistic |
| Intercept                                     | −0.0021     |        | −973                |
| <i>OIB#<sub>t−1</sub></i>                     | 0.0426      |        | 30.75               |
| Adjusted <i>R</i> <sup>2</sup>                |             | 0.0049 |                     |
| Intercept                                     | −0.0010     |        | −4.83               |
| <i>OIB\$<sub>t−1</sub></i>                    | 0.0259      |        | 31.40               |
| Adjusted <i>R</i> <sup>2</sup>                |             | 0.0051 |                     |

five-minute returns by about 0.1%, which is substantial, given the time interval (five minutes). The explanatory power of the regression is small, however, with the *R*<sup>2</sup> being only about 0.5%.

To gain insight regarding how market efficiency has changed over time, predictive regressions identical to those in Table 3 are computed for each calendar month in the ten-year sample. Fig. 1 plots the resulting *R*<sup>2</sup>'s and *t*-statistics for lagged *OIB\$*. Clearly, the degree of NYSE market efficiency has improved over this decade. The *R*<sup>2</sup> has moved from a peak of about 8% in the eighths regime to virtually zero by the end of the sample period. The *t*-statistic on imbalance has also gone from around 12 at the start of the sample to considerably less than two towards the end.

We now turn to the interaction of day-to-day liquidity with market efficiency. For parsimony, we focus on dollar imbalances (*OIB\$*) and the effective spread (*ESPR*) measure of illiquidity.<sup>14</sup> These are probably the more informative choices because *OIB\$* provides the economic magnitude of the order imbalance, and the

<sup>14</sup>Using *OIB#* yields similar conclusions.

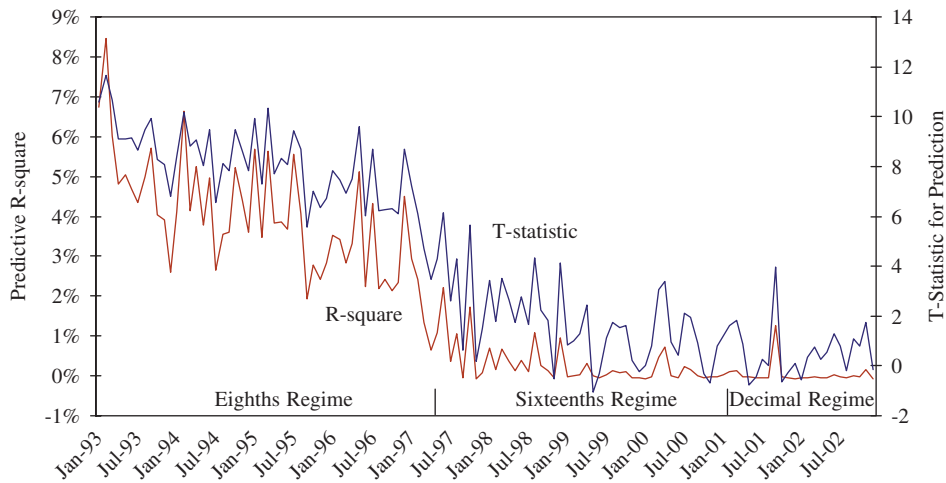


Fig. 1. Market inefficiency trend, NYSE, 1993–2002. Five-minute return predictions using lagged (by five minutes) dollar order imbalance.

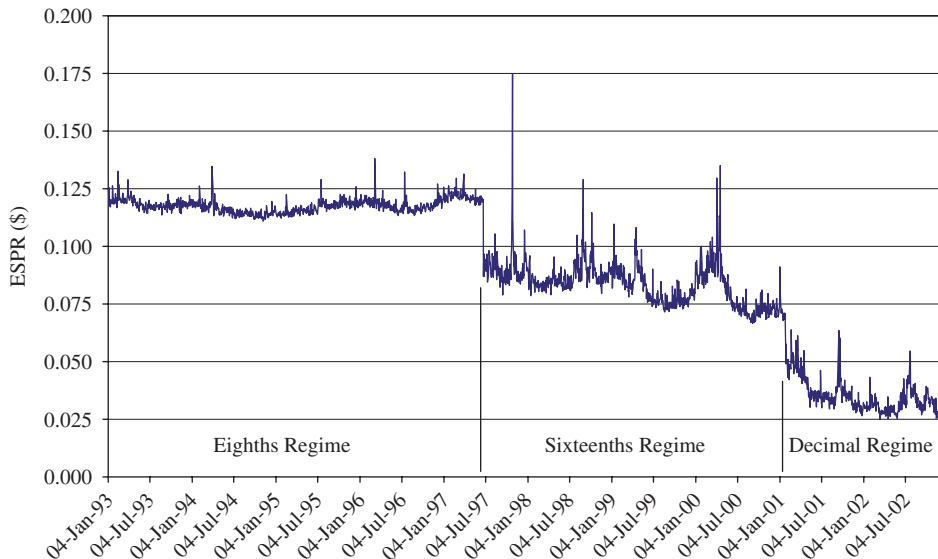


Fig. 2. Value-weighted daily average effective spread, NYSE, 1993–2002.

effective spread is closer to the actual transaction costs incurred by traders. Fig. 2 plots the time series of the effective spread (*ESPR*). Clearly, *ESPR* has experienced three distinct regimes corresponding to subperiods for the eighth, sixteenth, and decimal minimum tick sizes. Moreover, *ESPR* appears to be trending to some extent within each subperiod. In order to examine the interaction of liquidity with return predictability, we need to separate liquid periods from illiquid ones within each regime. Based on analyses by Blume, MacKinlay, and Terker (1989), and Cox and Peterson (1994), and on experiences with specific events such as the LTCM crisis, we surmise that the effect of illiquidity on trading activity (and arbitrage activity in particular) is likely to be particularly pronounced during days of abnormally low liquidity. We therefore define “low liquidity” days as those when the linearly detrended effective spread for that regime is at least one standard deviation above the mean.

Table 4 reports summary statistics for effective spreads on low and high liquidity days. The numbers of illiquid days as a percentage of the total number of trading days are 16%, 13%, and 19%, in the eighth,



Table 4

## Distribution of liquid and illiquid days

Effective spreads (*ESPR*) are obtained from the NYSE TAQ database. Spreads are averaged across the day for each stock and then value weighted across stocks using market capitalizations at the beginning of each calendar year. Illiquid days are those on which the daily effective spread is at least one standard deviation above the detrended expected effective spread for the trading day. The detrending is done separately for each tick size regime. The sample spans the years 1993–2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993–June 23, 1997, the sixteenths June 24, 1997–January 28, 2001, and the decimal regime spans the remainder of the sample period.

|                   |                    | <i>ESPR</i> on liquid days | <i>ESPR</i> on illiquid days |
|-------------------|--------------------|----------------------------|------------------------------|
| Eighths regime    | Mean               | 0.1168                     | 0.1227                       |
|                   | Median             | 0.1170                     | 0.1219                       |
|                   | Standard Deviation | 0.0022                     | 0.0029                       |
|                   | Number of days     | 976                        | 154                          |
| Sixteenths regime | Mean               | 0.0819                     | 0.0986                       |
|                   | Median             | 0.0833                     | 0.0962                       |
|                   | Standard Deviation | 0.0072                     | 0.0117                       |
|                   | Number of days     | 804                        | 104                          |
| Decimal regime    | Mean               | 0.0328                     | 0.0462                       |
|                   | Median             | 0.0321                     | 0.0480                       |
|                   | Standard Deviation | 0.0049                     | 0.0077                       |
|                   | Number of days     | 404                        | 77                           |

sixteenth, and decimal regimes, respectively, and these proportions are reasonably close to each other. The mean effective spread on illiquid days is 120% of that on liquid days in the sixteenth regime and the corresponding ratio is 141% for the decimal regime. However, this ratio is only about 105% for the eighths regime, possibly due to the fact that the larger tick sizes may have caused the measured effective spread to be higher than its true value due to the impact of rounding,<sup>15</sup> leaving less scope for a significant upward movement of the spread on illiquid days relative to normal days.

To estimate the influence of liquidity on market efficiency, a low-liquidity dummy (*ILD*) is interacted with the basic explanatory variable (lagged *OIB\$*) already used in the regressions of Table 3. Thus, the interaction variable,  $OIB\$_{t-1} * ILD$ , equals  $OIB\$_{t-1}$  on days with low liquidity and zero otherwise. Table 5 presents the results of regressions that include the interaction variable.

For the portfolio that aggregates all sample firms, the coefficients on  $OIB\$_{t-1} * ILD$  are positive and significant in all subperiods, and those on  $OIB\$_{t-1}$  are significant in the two earlier regimes. The coefficient estimates of  $OIB\$_{t-1}$  have decreased steadily from 0.030 in the eighths regime to 0.006 in the decimal regime. The significantly positive coefficients on  $OIB\$_{t-1} * ILD$  suggest that the ability of lagged *OIB\$* to predict returns increases during periods of illiquidity. Thus, liquidity enhances efficiency. Indeed, during the sixteenth and decimal subperiods, the coefficients on  $OIB\$_{t-1} * ILD$  are more than twice as large as those on  $OIB\$_{t-1}$  alone. Hence, in these later subperiods, which are generally more efficient on average, illiquidity has a relatively stronger inhibiting influence. For instance, during the eighths regime a one-standard deviation increase in lagged *OIB\$* causes a 0.6% increase in returns when markets are liquid and a 0.8% increase in returns when markets are illiquid. During the decimal regime the impact on returns of a one standard deviation increase in lagged *OIB\$* is 0.1% during liquid periods and 0.4% during illiquid periods.

A decline in the significance and explanatory power of the regressions in Table 5 accompanies the general improvement in liquidity over the sample period. Adjusted  $R^2$ 's drop across the three tick size regimes from around 4% in the eighths regime to less than 1% in the decimal regime. Overall, the evidence suggests that the secular changes in spreads accompanying tick size reductions (Bessembinder, 2003) have been accompanied by a marked increase in the degree of market efficiency.

<sup>15</sup>See Ball and Chordia (2001).

Table 5

Predictive regressions of five-minute returns on lagged order imbalances, and lagged order imbalances interacted with a dummy variable for liquidity regimes

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. The five-minute return,  $Return_t$ , is the dependent variable in all regressions; it is computed using the mid-points of the first and last quotes within five-minute trading interval  $t$ .  $OIB\#_{t-1}$  is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during five-minute trading interval  $t-1$ .  $OIB\$_{t-1}$  is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during five-minute trading interval  $t-1$ . The illiquidity dummy,  $ILD$ , is 1.0 when the daily effective spread is at least one standard deviation above the detrended expected effective spread for the trading day, otherwise zero. The sample spans the years 1993 to 2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993 to June 23, 1997, the sixteenths June 24, 1997 to January 28, 2001, and the decimal regime spans the remainder of the sample period. The sample size is represented by  $n$ . (The first interval of each trading day is used only in the lagged regressors.)

|                                    |                     | Coefficient | <i>t</i> -Statistic |
|------------------------------------|---------------------|-------------|---------------------|
| Eighths Regime ( $n = 86,692$ )    | $OIB\$_{t-1}$       | 0.0298      | 45.53               |
|                                    | $OIB\$_{t-1} * ILD$ | 0.0241      | 13.64               |
|                                    | Intercept           | −0.0010     | −5.62               |
|                                    | Adjusted $R^2$      |             | 0.0348              |
| Sixteenths Regime ( $n = 69,538$ ) | $OIB\$_{t-1}$       | 0.0148      | 7.92                |
|                                    | $OIB\$_{t-1} * ILD$ | 0.0517      | 9.53                |
|                                    | Intercept           | −0.0012     | −2.99               |
|                                    | Adjusted $R^2$      |             | 0.0032              |
| Decimal Regime ( $n = 36698$ )     | $OIB\$_{t-1}$       | 0.0059      | 1.55                |
|                                    | $OIB\$_{t-1} * ILD$ | 0.0278      | 3.23                |
|                                    | Intercept           | −0.0003     | −0.35               |
|                                    | Adjusted $R^2$      |             | 0.0005              |

### 3.2. Size-based subsamples

It is of interest to explore patterns in the liquidity-efficiency relation across groups stratified by firm size. Table 6 presents results for subsamples formed by ranking firms by market capitalization at the beginning of the year, dividing the ranked firms into thirds, and then calculating value-weighted returns, imbalances, and detrended effective spreads within each tierce. Again, the coefficients on  $OIB\$_{t-1}$  and  $OIB\$_{t-1} * ILD$  are positive and significant in eight of the nine cases corresponding to the three firm size groups and the three tick size regimes. The only exception is that for the large firms during the decimal regime, when only the interactive variable is significant.

The coefficient estimates on lagged  $OIB\$$  across the size groups for the eighths regime decline from 0.029 for the large firms to 0.012 for the small firms.<sup>16</sup> Across the tick size regimes, the coefficients for the large firms show a monotonic decline with the tick size, as compared to the smaller firms. The coefficient for the smaller size groups relative to those for the large caps is about two to ten times larger in the sixteenths and decimal regimes. The coefficient patterns suggest that the efficiency of the large cap sector has increased with the reduction in minimum tick size and also imply that in the smaller tick size regimes, the degree of market efficiency is greater for larger firms. Perhaps this finding is explained by the eighth tick size having been more binding on the larger firms (Ball and Chordia, 2001). Thus, the greater decrease in spreads for large firms accompanying tick size reductions may account for increased market efficiency in the decimal regimes for these firms.

<sup>16</sup>An interesting aspect of the regressions is that during the eighths regime the coefficient of lagged imbalance alone is *greater* for larger firms. This highlights the issues arising from cross-sectional comparisons. For example, Chordia and Subrahmanyam (2004) find that imbalance autocorrelations are *greater* for larger firms (they attribute this finding to greater influence of institutional herding in large firms). Since autocorrelated imbalances are the key driver of return forecastability in Chordia and Subrahmanyam's (2004) setting, stronger serial correlations in imbalances in their model imply stronger return predictability from order flows. Thus, the higher imbalance autocorrelations for larger firms may explain why the lagged imbalance coefficients in the return regressions are larger for such firms.

Table 6

Predictive regressions of five-minute returns on lagged order imbalances, and lagged order imbalances interacted with a dummy variable for liquidity regimes, by firm-size

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. The five-minute return,  $Return_t$ , is the dependent variable in all regressions; it is computed using the midpoints of the first and last quotes within five-minute trading interval  $t$ .  $OIB\#_{t-1}$  is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during five-minute trading interval  $t-1$ .  $OIB\$_{t-1}$  is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during five-minute trading interval  $t-1$ . The illiquidity dummy, *ILD*, is 1.0 when the daily effective spread is at least one standard deviation above the detrended expected effective spread for the trading day, otherwise zero. The sample spans the years 1993–2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993–June 23, 1997, the sixteenths June 24, 1997–January 28, 2001, and the decimal regime spans the remainder of the sample period. Each firm size category includes one-third of all sample firms. The first interval of each trading day is used only in the lagged regressors. The sample size is the number of available five-minute intervals in each subperiod, as reported in Table 1, less the number of days (the first interval of each trading day is used only in the lagged regressors).

|                   |                     | Large Firms |                     | Mid-Cap Firms |                     | Small Firms |                     |
|-------------------|---------------------|-------------|---------------------|---------------|---------------------|-------------|---------------------|
|                   |                     | Coefficient | <i>t</i> -Statistic | Coefficient   | <i>t</i> -Statistic | Coefficient | <i>t</i> -Statistic |
| Eighths Regime    | $OIB\$_{t-1}$       | 0.0293      | 41.47               | 0.0192        | 37.79               | 0.0115      | 32.29               |
|                   | $OIB\$_{t-1} * ILD$ | 0.0222      | 11.15               | 0.0097        | 7.58                | 0.0103      | 11.39               |
|                   | Intercept           | −0.0010     | −4.68               | −0.0006       | −4.12               | 0.0000      | −0.21               |
|                   | Adjusted $R^2$      | 0.0277      |                     | 0.0227        |                     | 0.0194      |                     |
| Sixteenths Regime | $OIB\$_{t-1}$       | 0.0094      | 4.63                | 0.0228        | 17.67               | 0.0207      | 21.39               |
|                   | $OIB\$_{t-1} * ILD$ | 0.0514      | 8.62                | 0.0299        | 8.68                | 0.0318      | 12.37               |
|                   | Intercept           | −0.0008     | −1.82               | −0.0019       | −6.15               | −0.0015     | −5.79               |
|                   | Adjusted $R^2$      | 0.0019      |                     | 0.0079        |                     | 0.0129      |                     |
| Decimal Regime    | $OIB\$_{t-1}$       | 0.0029      | 0.74                | 0.0111        | 3.44                | 0.0245      | 8.84                |
|                   | $OIB\$_{t-1} * ILD$ | 0.0287      | 3.11                | 0.0187        | 2.74                | 0.0259      | 4.45                |
|                   | Intercept           | −0.0002     | −0.20               | −0.0005       | −0.82               | −0.0022     | −3.63               |
|                   | Adjusted $R^2$      | 0.0003      |                     | 0.0007        |                     | 0.0039      |                     |

### 3.3. Robustness checks

We now present the results of additional analyses that are intended to test the interpretation and reliability of our results.

#### 3.3.1. Issues surrounding interpretation of results

In interpreting the interaction coefficient on *OIB* and illiquidity, one possible issue is that an exogenous shock might cause extreme order imbalances and simultaneously reduce liquidity. We could be picking up the effect of the shock rather than capturing the role of liquidity in assisting the establishment of market efficiency. To investigate this possibility, we compute absolute order imbalances within liquid and illiquid periods for each of the three size groups, across the tick size regimes. In each of the nine cases, the absolute imbalances differed only minimally across liquid and illiquid periods (the difference is less than 1% in every case). This finding suggests that the identified illiquid periods are not capturing periods of abnormally high (absolute) order imbalances. Moreover, there is no reason to believe that illiquidity is jointly determined with *signed* imbalances. Thus, when markets are illiquid, one may expect to see fewer buy as well as fewer sell orders, but this has no implication for net order flow, which is our right-hand variable.

As a further check against an unknown contemporaneous influence on return predictability and illiquidity, we try an alternative definition of illiquid periods based on the effective spread on the previous day (as opposed to the contemporaneous spread), and find that this does not have any material impact upon the central results. This result is to be expected, given the high autocorrelation in liquidity documented by Chordia, Sarkar, and Subrahmanyam (2005).

To shed additional light on the finding that increased liquidity on a given day is associated with more efficient markets on the next day, we conduct Granger causality tests, which relate our efficiency metric to

illiquidity. In this process, we first perform regressions of returns on lagged imbalances on a daily basis. Specifically, we estimate one time-series regression of the type in Table 3 for each day in our sample, using all five-minute intervals available for that day. Next, we used the imbalance coefficient and our illiquidity measure (i.e., the effective spread) in a vector autoregression. Thus, we estimate a bivariate system in which the daily imbalance coefficient and the daily effective spread are each modeled as linear functions of lagged coefficients and lagged values of the spread.<sup>17</sup> We then perform our causality tests using the resulting coefficient estimates. We find that the  $p$ -value for the null that the effective spread Granger-causes the lagged imbalance coefficient is 0.001, whereas that for the reverse hypothesis that the coefficient causes the spread is 0.141. The respective  $p$ -values for the (former, latter) causality tests within tick size regimes were (0.073, 0.202), (0.039, 0.105), and (0.032, 0.417) for the eighth, sixteenth, and decimal subperiods, respectively. Thus, in each case there is evidence that the effective spread Granger-causes the coefficient in the regression of returns on lagged imbalance, but there is no evidence of econometric causality running from the regression coefficient to the spread. This finding supports our interpretation that liquidity causes markets to be more efficient.

Since previous work (Chordia and Subrahmanyam, 2004) suggests that autocorrelated imbalances may be a key source of predictability of future returns from imbalances, one concern is that our results may be influenced by the behavior of imbalance autocorrelations over time. It is therefore worth considering how imbalance autocorrelations have behaved across the three tick size regimes. We find that the first-order serial correlations in  $OIB\$$  are 0.266, 0.186, and 0.187 during the eighths, sixteenths, and decimal regimes, respectively. While there is some decrease in autocorrelations during the smaller tick regimes, it is not enough to reduce the  $R^2$  from about 8% at the beginning of the sample (Fig. 1) to virtually zero in the decimal period. Also, from the sixteenth to the decimal period, the  $R^2$  and predictive  $t$ -statistic dropped further with no reduction in the autocorrelation of imbalance. The evidence thus supports the hypothesis that increased liquidity allows outsiders to more effectively assist the designated dealer (i.e., the specialist) in absorbing investor demands, which assists in the establishment of efficiency.

Our basic contention is that return predictability from order flows is diminished during liquid periods because market makers receive greater assistance from outsiders in these periods. Thus, we propose that return predictability from order flows arises because of the limited ability of rational market makers to absorb outsider demands, rather than due to any cognitive limitation on the part of market makers. To bolster this notion, we decompose order imbalances into expected and unexpected components using an AR(12) model for  $OIB$ .<sup>18</sup> We would expect rational market makers using Bayesian updating to largely incorporate the effects of predictable imbalances in prior periods, implying that the predictability of returns from lagged order flows should arise principally from the unexpected component of order flow in any given period. Results of predictive return regressions using the expected and unexpected components of  $OIB$  are presented in Table 7.<sup>19</sup>

In general, the results show that the predictability of returns emanates largely from the unexpected component of  $OIB$ . In almost every case, only the innovation coefficients are significant. While there is one instance in the eighths regime where the coefficient of the predictable component is significant, its magnitude is smaller than that of the corresponding innovation coefficient. Also, there is reason to believe that the significance of the predictable component may have arisen due to the large tick size, which may have caused a sluggish price response to order imbalances in the eighths regime. Overall, the evidence suggests that market makers react rationally to order flow and that in liquid periods they receive assistance from outsiders, which mitigates return predictability from order flow.

It also is worth obtaining some perspective on the role of institutional frictions, such as price continuity rules, that may preclude NYSE specialists from responding quickly to large imbalances and thus influence return predictability. In this regard, note that our central hypothesis is not that returns are predictable from order flows (this has been established in earlier literature, e.g., Chordia and Subrahmanyam, 2004). Rather, we

<sup>17</sup>We use the Akaike information criterion to select the lag length based on the corresponding vector autoregression. Use of the alternative Schwarz criterion leads to similar results. The lag length used ranges from two to eight across the four autoregressions for the full sample and the three tick size regimes.

<sup>18</sup>We try a number of alternative lag structures going from three to 15 lags, and find qualitatively similar results.

<sup>19</sup>In the remainder of this subsection and the next one, we revert to using the effective spread as a measure of illiquidity, though the results are qualitatively identical if quoted spreads are used.

Table 7

Predictive regressions of five-minute returns on lagged expected and unexpected order imbalances, and lagged order imbalances interacted with a dummy variable for liquidity regimes

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. The five-minute return,  $Return_t$ , is the dependent variable in all regressions; it is computed using the mid-points of the first and last quotes within five-minute trading interval  $t$ . *OIB* is measured as the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during five-minute trading interval  $t-1$ .

*OIB* is decomposed using an AR(12) model into unexpected (*OIBI*) and expected (*OIBP*) components. The illiquidity dummy, *ILD*, is 1.0 when the daily effective spread is at least one standard deviation above the detrended expected effective spread for the trading day, otherwise zero. The sample spans the years 1993 to 2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993–June 23, 1997, the sixteenths June 24, 1997–January 28, 2001, and the decimal regime spans the remainder of the sample period. The sample size is the number of available five-minute intervals in each sub-period, as reported in Table 1, less the number of days (the first interval of each trading day is used only in the lagged regressors).

|                   |                      | Coefficient | <i>t</i> -Statistic |
|-------------------|----------------------|-------------|---------------------|
| Eighths Regime    | $OIBI\$_{t-1}$       | 0.0311      | 39.27               |
|                   | $OIBI\$_{t-1} * ILD$ | 0.0237      | 10.87               |
|                   | $OIBP\$_{t-1}$       | 0.0158      | 5.90                |
|                   | $OIBP\$_{t-1} * ILD$ | 0.0038      | 0.58                |
|                   | Intercept            | −0.0000     | −0.12               |
|                   | Adjusted $R^2$       |             | 0.0257              |
| Sixteenths Regime | $OIBI\$_{t-1}$       | 0.0131      | 7.51                |
|                   | $OIBI\$_{t-1} * ILD$ | 0.0310      | 6.15                |
|                   | $OIBP\$_{t-1}$       | 0.0012      | 0.16                |
|                   | $OIBP\$_{t-1} * ILD$ | 0.0131      | 1.80                |
|                   | Intercept            | 0.0003      | 0.50                |
|                   | Adjusted $R^2$       |             | 0.0016              |
| Decimal Regime    | $OIBI\$_{t-1}$       | 0.0023      | 0.60                |
|                   | $OIBI\$_{t-1} * ILD$ | 0.0254      | 2.67                |
|                   | $OIBP\$_{t-1}$       | 0.0157      | 1.03                |
|                   | $OIBP\$_{t-1} * ILD$ | −0.0195     | −1.03               |
|                   | Intercept            | −0.0005     | −0.33               |
|                   | Adjusted $R^2$       |             | 0.0003              |

argue that liquidity enhances efficiency, a claim that would seem to be worth investigating even in the presence of price continuity rules. Nonetheless, one possible way to address this issue is to use NASDAQ stocks, which are not subject to price continuity rules. For these stocks, however, there are issues related to measurement of liquidity due to the lack of odd-eighths quotes in the earlier years of our sample period, which reduces the time-series variation in spreads (Christie and Schultz, 1994). Furthermore, many NASDAQ stocks are not actively traded, which further complicates any analysis of intraday efficiency. We address these issues by taking the largest ten NASDAQ stocks ranked by market capitalization at the end of 2000. We then apply our portfolio technique and run regressions analogous to those in Table 5 for the sample from January 29, 2001 to December 31, 2002 (the NYSE post-decimal period). The results are similar to those in Table 5; while order imbalance itself is not significant, its interaction with the illiquid days indicator is significant with a magnitude (*t*-statistic) of 0.259 (15.13), and both numbers are much larger than the corresponding ones in Table 5. This suggests that even in the absence of price continuity rules, liquidity enhances efficiency, and the effect appears to be stronger in NASDAQ stocks.

### 3.3.2. Alternative illiquidity specifications

Our liquidity indicator has hitherto been the effective spread, which measures the deviation of transaction prices from quote midpoints. In a sense, this measure is related to the contemporaneous price impact of a trade. While there is no reason to expect a mechanical relation between the contemporaneous price impact and future return predictability from past order imbalances, there may still be a residual concern about relating a transaction-price based liquidity measure to the return-imbalance relation. Thus, for robustness, we also try an alternative illiquidity indicator, namely, the quoted spread (*QSPR*). This measure, being the difference

between quoted ask and quoted bid prices, is not mechanically related to the transaction price response to order flow. We calculate the variable *ILD* based on *QSPR* and run regressions analogous to those in Tables 5 and 6. The results are qualitatively unaltered relative to those in the earlier tables. In particular, return predictability is diminished during liquid days in every instance. This lends confidence that our previous results are robust to alternative measures of liquidity.

Another robustness check involves examining alternative procedures for computing illiquidity innovations. Our previous detrending procedure is a parsimonious but subjective choice. We examine an alternative procedure whereby the effective spread is regressed on linear as well as quadratic trend terms. In addition, following Roll, Schwartz, and Subrahmanyam (2007), we include indicator variables for calendar effects, such as day-of-the-week and month-of-the-year regularities, and for trading days that precede major holidays (Christmas, New Year's Day, Independence Day, and Thanksgiving). Illiquidity innovations are defined as the residual from this augmented regression. Note that adjustment for the preceding variables can potentially work against finding support for our main thesis. This is because markets may be more efficient during days of the week when the market is more liquid, and adjusting for calendar regularities would negate this effect. Nonetheless, the results from this alternative adjustment procedure, presented in Table 8, show that the significance of the variables improves slightly, but the results are otherwise unchanged, indicating that previous results are robust to the illiquidity adjustment procedure.

### 3.3.3. Individual stock regressions

Our final check addresses the fact that we use portfolios rather than individual stocks (as do Brennan, Jagadeesh, and Swaminathan, 1993; Lo and MacKinlay, 1990, in their studies of market efficiency). To examine the impact of the portfolio method, we also calculate regressions for individual stocks using illiquid days defined relative to the effective spread for each stock. The regressions are analogous to those in Table 5, but we also control for contemporaneous and lagged market return to reduce cross-sectional correlations in

Table 8

Predictive regressions of five-minute returns on lagged order imbalances, and lagged order imbalances interacted with a dummy variable for liquidity regimes, using an expanded model of illiquidity innovations

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. The five-minute return,  $Return_t$ , is the dependent variable in all regressions; it is computed using the mid-points of the first and last quotes within five-minute trading interval  $t$ .  $OIB\#_{t-1}$  is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during five-minute trading interval  $t-1$ .  $OIB\$_{t-1}$  is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during five-minute trading interval  $t-1$ . The illiquidity dummy, *ILD*, is 1.0 when the daily effective spread is at least one standard deviation above the adjusted expected effective spread for the trading day, otherwise zero. The adjustment consists of regressing effective spreads on day of the week, month of the year, linear and quadratic trend terms, and a dummy variable for whether a day preceded a holiday (Independence Day, Christmas Day, or Thanksgiving). The sample spans the years 1993 to 2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993 to June 23, 1997, the sixteenths June 24, 1997 to January 28, 2001, and the decimal regime spans the remainder of the sample period. The sample size is the number of available five-minute intervals in each sub-period, as reported in Table 1, less the number of days (the first interval of each trading day is used only in the lagged regressors).

|                   |                     | Coefficient | <i>t</i> -statistic |
|-------------------|---------------------|-------------|---------------------|
| Eighths Regime    | $OIB\$_{t-1}$       | 0.0304      | 46.46               |
|                   | $OIB\$_{t-1} * ILD$ | 0.0195      | 11.04               |
|                   | Intercept           | -0.0010     | -5.67               |
|                   | Adjusted $R^2$      | 0.0341      |                     |
| Sixteenths Regime | $OIB\$_{t-1}$       | 0.0138      | 7.38                |
|                   | $OIB\$_{t-1} * ILD$ | 0.0577      | 10.79               |
|                   | Intercept           | -0.0011     | -2.80               |
|                   | Adjusted $R^2$      | 0.0035      |                     |
| Decimal Regime    | $OIB\$_{t-1}$       | 0.0059      | 1.59                |
|                   | $OIB\$_{t-1} * ILD$ | 0.0330      | 3.55                |
|                   | Intercept           | -0.0003     | -0.36               |
|                   | Adjusted $R^2$      | 0.0005      |                     |



Table 9

Individual Stock Predictive regressions of five-minute returns on lagged order imbalances, and lagged order imbalances interacted with a dummy variable for liquidity regimes

Returns and order imbalances (*OIB*) are from the NYSE TAQ database. The five-minute return,  $Return_t$ , is the dependent variable in all regressions; it is computed using the mid-points of the first and last quotes within five-minute trading interval  $t$ .  $OIB\#_{t-1}$  is the number of buyer-initiated less the number of seller-initiated trades divided by the total number of trades during five-minute trading interval  $t-1$ .  $OIB\$_{t-1}$  is the total dollars paid by buyer-initiators less the total dollars received by seller-initiators divided by the dollar volume of trading during five-minute trading interval  $t-1$ . The illiquidity dummy, *ILD*, is 1.0 when the daily effective spread of a stock is at least one standard deviation above the adjusted expected effective spread for the trading day, otherwise zero. The sample spans the years 1993–2002 inclusive and consists of all stocks that traded every day on the NYSE during the sample period. The regression includes the contemporaneous and lagged equally-weighted average of all sample returns (the coefficients of these variables are not reported for brevity). The eighths regime spans January 1, 1993–June 23, 1997, the sixteenths June 24, 1997–January 28, 2001, and the decimal regime spans the remainder of the sample period. The first interval of each trading day is used only in the lagged regressors. The  $t$ -statistics are adjusted for cross-correlations in the residuals. All coefficients are multiplied by 1000.

|                   |                        | Coefficient | $t$ -statistic |
|-------------------|------------------------|-------------|----------------|
| Eighths Regime    | $OIB\$_{t-1}$          | 0.1369      | 54.04          |
|                   | $OIB\$_{t-1} * ILD$    | 0.0318      | 6.17           |
|                   | Intercept              | 0.0041      | 8.26           |
|                   | Average adjusted $R^2$ | 0.389       |                |
| Sixteenths Regime | $OIB\$_{t-1}$          | 0.1130      | 31.69          |
|                   | $OIB\$_{t-1} * ILD$    | 0.0471      | 5.42           |
|                   | Intercept              | −0.0024     | −4.31          |
|                   | Adjusted $R^2$         | 0.333       |                |
| Decimal Regime    | $OIB\$_{t-1}$          | −0.0154     | −1.62          |
|                   | $OIB\$_{t-1} * ILD$    | 0.0518      | 2.65           |
|                   | Intercept              | 0.0006      | −0.35          |
|                   | Adjusted $R^2$         | 0.367       |                |

the regression residuals.<sup>20</sup> We also account for any remaining cross-correlation by correcting the  $t$ -statistic of the mean coefficient across all individual stock regressions. This correction, mentioned in Footnote 8 of Chordia, Roll, and Subrahmanyam (2000), assumes that the residual cross-correlation and the residual variance are homogeneous across stocks. Under those assumptions, the standard error of the cross-sectional mean is inflated by  $[1+(N-1)\rho]^{1/2}$ , where  $N$  is the number of regressions and  $\rho$  is the common cross-correlation in the residuals,<sup>21</sup> which is proxied by the average residual cross-correlation across all pairs of the individual stock regressions. The cross-sectional average regression coefficients are presented in Table 9. As can be seen, the significance pattern of the coefficients mimics that in Table 5, though the magnitude of the coefficients is smaller. Again, results for the size-stratified portfolios are not presented for brevity but are broadly consistent with those in Table 6.

Overall, based on all of the evidence, the evidence largely supports the hypothesis that efficiency-creating activity is enhanced when the market is liquid. In the next section, we show how variance ratios can be used to shed further light on the relation between liquidity and efficiency.

#### 4. Variance ratios

##### 4.1. Deviations from a random walk benchmark across the tick size regimes

The preceding analysis focuses on the predictive relation between returns and order flows intraday as an inverse measure of market efficiency and, implicitly, the quality of a financial market. An alternative approach

<sup>20</sup>The market return is defined simply as the equally-weighted average of returns across all stocks within a particular five minute interval.

<sup>21</sup>The formula of Chordia, Roll, and Subrahmanyam (2000) contains a typographical error. Specifically, it contains an extraneous digit, 2.

involves the comparison of short- and long-horizon variance ratios. This approach is based on the observation that for a random walk price process, the variance of long-horizon returns is  $q$  times the variance of short horizon returns, where  $q$  is the number of short horizon intervals in the longer horizon.<sup>22</sup> Deviations from a random walk (efficient-market) benchmark can arise because inventory control activity can induce return serial correlation (Grossman and Miller, 1988). But the deviation would be mitigated if arbitrageurs assist the market maker in absorbing outside orders, and our hypothesis contends that there should be smaller deviations from random walk benchmarks during more liquid regimes when trading is less costly.

We consider the ratio of midquote return variances computed from five-minute intervals and from open-to-close of trading days. In computing this variance ratio, the five-minute return variance is multiplied by the number of five-minute intervals in a trading day. For a random walk, this scaled variance ratio would converge to unity in large samples.

Panel A of Table 10 reports the five-minute/open-close variance ratios for each tick size regime and for each size-based portfolio. Consistent with intuition, the variance ratios are closer to unity for larger firms, indicating that the prices for these firms conform more closely to random walks. The point estimates of the variance ratios are lowest for the decimal regimes. In addition, the variance ratios for small and midcap firms in the decimal regime are significantly lower relative to the corresponding ones in the eighths regime. This indicates that the extent of “noise” induced by the trading process is smaller in the decimal regime, especially for the small and midcap firms. Given the secular decrease in bid-ask spreads brought about by the reduction in the tick size, the evidence accords with our hypothesis that deviations from random walk benchmarks should be reduced when markets are more liquid.

#### 4.2. Open-close/close-open return variance ratios and return autocorrelations

The preceding analysis implicitly assumes deviations from a random walk benchmark to be a metric for the noise introduced by the trading process. Lower levels of such noise implicitly imply higher market quality. However, the microstructure literature also considers the degree to which prices reflect private information about fundamentals as an indicator of a financial market’s quality. This measure is generally termed informational efficiency. Thus, even if returns are not predictable from past *public* information, they can reflect varying degrees of *private* information, as implied by Kyle’s (1985) model.

It is reasonable to expect more liquid markets to be accompanied by greater informational efficiency. For example, if a smaller tick size allows agents to trade on smaller pieces of information, one would expect more informed trading as the tick size decreases. Hence, informational efficiency should progressively increase across the three tick size regimes. This is not the only possible hypothesis, however. If a reduction in tick sizes adversely affects institutional execution costs due to decreased depth (Jones and Lipson, 2001), informational efficiency could be adversely affected because of reduced trading on information by large institutions.

We use the ratios of open-to-close and close-to-open return variances to provide some evidence regarding these hypotheses. French and Roll (1986) relate these ratios to the amount of information incorporated into prices. They show that (per hour) open-close return variances are greater than (per hour) close-open ones and offer three potential explanations for this finding: (i) incorporation of private information during trading hours, (ii) mispricing caused by investor misreaction or market frictions and microstructure noise induced by bid-ask bounce, and (iii) greater incorporation of public information into prices during trading hours. They reject (iii) because the variance ratios are not significantly different on business days when the stock market is closed. They conclude that the other two components help explain the higher ratio during market trading hours, with (i) being the dominant factor. Relying on their conclusions, the variance ratio measures either mispricing or the incorporation of private information into prices during trading hours. Thus, the variance ratio is possibly linked to the informativeness or efficiency of the pricing system in the sense of Kyle (1985).

We seek to discern whether variance ratios have changed across the three tick size regimes, and if so, whether the pattern of changes reveals which of the two components identified by French and Roll (1986) is

<sup>22</sup>See, for example, Lo and MacKinlay (1990).

Table 10

## Daily variance ratios and autocorrelations

Panel A presents the ratio of five minute return variance to open-to-close return variance (scaled by the number of five minute intervals in a day). Panel B presents (open-to-close) ÷ (close-to-open) per hour return variance ratios, based on mid-quote returns, across three tick size regimes (eighths, sixteenths, and decimals). Panel C presents daily first order return autocorrelations across these regimes. The sample spans the years 1993 to 2002 inclusive, and consists of all stocks that traded every day on the NYSE during the sample period. The eighths regime spans January 1, 1993 to June 23, 1997, the sixteenths June 24, 1997 to January 28, 2001, and the decimal regime spans the remainder of the sample period. In Panel C, *p*-values for the relevant numbers being statistically different from zero are provided in parentheses. A \* (\*\*) in Panel A (Panel B) indicates that the relevant number is statistically higher than the corresponding number in the eighths (sixteenths) regime at the 5% level. A \* (†) in Panels A and C indicates that the relevant number is statistically different from the corresponding number in the eighths regime at the 5% (10%) level. The statistical conclusions for Panels A and B derive from bootstrap analyses described in the text.

|                                                               | Eighths           | Sixteenths         | Decimals           |
|---------------------------------------------------------------|-------------------|--------------------|--------------------|
| <i>Panel A: Five minutes/daily variance ratios</i>            |                   |                    |                    |
| Large firms                                                   | 1.21              | 1.21               | 1.10               |
| Mid-cap firms                                                 | 1.81              | 1.82               | 1.36*              |
| Small firms                                                   | 2.28              | 2.38               | 1.79†              |
| <i>Panel B: Per hour open/close variance ratios</i>           |                   |                    |                    |
| Large firms                                                   | 6.17              | 7.14               | 6.69               |
| Mid-cap firms                                                 | 11.85             | 7.88               | 18.01**            |
| Small firms                                                   | 8.95              | 15.33              | 22.16*             |
| <i>Panel C: First order autocorrelations of daily returns</i> |                   |                    |                    |
| Large firms                                                   | 0.0572<br>(0.055) | −0.0228<br>(0.493) | 0.0039<br>(0.932)  |
| Mid-cap firms                                                 | 0.0924<br>(0.002) | 0.0697<br>(0.036)  | 0.0533<br>(0.244)  |
| Small firms                                                   | 0.2121<br>(0.000) | 0.0814*<br>(0.014) | 0.0339*<br>(0.458) |

relatively more important.<sup>23</sup> Panel B of Table 10 presents the (open-close) ÷ (close-open) per hour variance ratios for the three tick size regimes and the three size groups. To obtain per hour variances, open-close and close-open returns are first used separately to compute raw variances. Then, each computed variance is divided by the total number of calendar hours over the subsample in question.

As shown in the table, all per hour variance ratios are much higher than unity. To test statistical significance, we compute a nonparametric sign test from paired open-close and close-open squared returns, each normalized by the number of hours in their respective periods. The computed *t*-statistics far exceed two over the nine comparisons corresponding to the variance ratios in Table 10, thereby indicating a very high level of statistical probability that open-close volatility far exceeds close-open volatility. This finding is consistent with that of French and Roll (1986).

Table 10 also reveals that all of the variance ratios in the decimal regime are greater than the corresponding ratios in the eighths regime. We employ a bootstrap analysis to check the statistical significance of these differences. In the bootstrap, pseudo regimes, with the same numbers of observations as in the original data, are formed by randomly sorting actual dates, with replacement, and then by computing a ratio of ratios, i.e., (open-close) ÷ (close-open) variance ratio in a lower pseudo tick size regime divided by the (open-close) ÷ (close-open) variance ratio in the pseudo eighths regime. This procedure is repeated 5000 times, and the actual observed ratio of the variance ratios is compared to the relevant fractile of the bootstrapped distribution. The tests indicate that the per hour (open-close) ÷ (close-open) variance ratios for midcap firms

<sup>23</sup>We also estimate variance ratios across liquid and illiquid days (defined as in the previous section) and find that return variances are consistently higher on illiquid days. On the one hand, this is consistent with the notion that high illiquidity increases asset price fluctuations (Subrahmanyam, 1994). However, there is an endogeneity problem because exogenous volatility shocks (unrelated to private information) could cause reduced liquidity due to greater inventory risk. Because of the resulting interpretational problem, we do not present details of these empirical results.

during the decimal regime are statistically greater than the corresponding ratios in the sixteenth regime; the same conclusion applies for smallcap firms in the decimal regime vis-à-vis the eighths regime. The point estimates indicate that the variance ratios for the small and midcap firms generally increased in the decimal period relative to the other periods.

As mentioned earlier the higher variance ratios can be attributed either to an increase in mispricing or to an increase in privately informed trading that results in the incorporation of more information into prices when the market is open. To provide some suggestive evidence that distinguishes between these two possibilities, we consider the first-order daily return autocorrelations (based on the midpoint of the last quote in effect at the end of a trading day) across the three tick regimes. French and Roll (1986) argue that the absolute levels of these autocorrelations should be positively related to microstructural frictions or misreactions of investors to new information, and thus capture mispricing.<sup>24</sup>

The autocorrelations are presented in Panel C of Table 10 (together with *p*-values for whether they are statistically different from zero). They are positive in all but one (statistically insignificant) case, and they generally decrease across the regimes. For example, none of the autocorrelations is significant in the decimal regime, whereas five of the six autocorrelations in the other two regimes are positive and significant. Further, the lower autocorrelations in the sixteenth and decimal regimes for the smallest firms are statistically different from the corresponding one in the eighths regime. In considering these autocorrelations, note that nonsynchronous trading and bid-ask bounce are not materially relevant in our context, because firms are included only if they traded every day and end-of-day midquotes are used to compute returns. The positive autocorrelations are thus consistent with investor underreaction to information (Barberis, Shleifer, and Vishny, 1998), and/or with incomplete adjustment to autocorrelated order flow owing to market maker risk aversion (Chordia and Subrahmanyam, 2004). Either of these two phenomena can be termed “mispricing,” in the sense of French and Roll (1986). Such mispricing may arise due to behavioral biases (the former case) or insufficient market making capacity (the latter case).

Regardless of the cause, if such mispricing were driving the increase in variance ratios across time, autocorrelations should have increased along with variance ratios as the tick size decreased; but there is no evidence of this increase. In fact, there is reliable evidence that the opposite transpired for smaller firms. Consequently, the decrease in the variance ratios is consistent with the more effective incorporation of private information into prices in the lower tick regimes, especially for smaller firms. A plausible rationale for this phenomenon is that with lower bid-ask spreads, informed traders might find it worthwhile to trade on private information even with modest profit potential.<sup>25</sup>

Overall, all of the evidence is consistent with our proposed hypotheses. Increases in liquidity facilitate efficiency via two distinct channels. First, return predictability from order flows is diminished during periods of high liquidity because arbitrageurs are better able to assist specialists in absorbing order flows during such periods. Second, a reduction in the minimum price change allows for the collection of more information that, in turn, increases informational efficiency by allowing prices to reflect more information about fundamentals.

## 5. Conclusions

In an efficient market, return predictability from past information should be short-lived and minimal. Given the evidence that such predictability does exist in the short run, understanding its time variation and its relation to other financial market attributes, such as liquidity, are of fundamental importance. Based on this motivation, we examine how the predictive relation between returns and order flow varies over time and across different liquidity regimes.

<sup>24</sup>In Kyle (1985), where market makers have unlimited market making capacity, autocorrelations are zero. However, nonzero autocorrelations obtain in irrational investor models such as Daniel, Hirshleifer, and Subrahmanyam (1998) as well as in the inventory model of Grossman and Miller (1988).

<sup>25</sup>The increased informed trading, in turn, could potentially reduce equilibrium liquidity (increase spreads) in the post-decimal period. Our variance ratio results in Table 10, coupled with the finding that actual spreads generally decreased over time (Fig. 2), suggest that while decreased tick sizes have allowed for increased informed trading, the effect of this increase on illiquidity has not been sufficient to cause spreads to rise to the levels during the eighths regime.

Our analysis relies on a continuous series of short-horizon returns for a comprehensive sample of all NYSE stocks that traded every day during the ten-year period from 1993 to 2002. Return predictability from order flows has declined substantially over time with reductions in the minimum tick size. Such predictability is markedly diminished during liquid periods within each tick regime. Prices are closer to random walk benchmarks during the more recent decimal tick size regime than in earlier ones. The overall evidence is consistent with the hypothesis that increased arbitrage activity during more liquid periods enhances market efficiency.

Distinct from the Fama (1970) notion that efficiency implies a lack of return predictability, the microstructure literature also considers informational efficiency, which is defined as the amount of private information revealed in prices. We shed light on this measure of a financial market's quality by considering patterns in per hour open-to-close/close-to-open variance ratios. Variance ratios generally have increased while first-order return autocorrelations have declined as the minimum tick size was reduced; this pattern is particularly strong for smaller firms. This suggests that the observed increase in variance ratios is not due to increased mispricing, but to more private information being reflected in prices following the tick size reductions. In sum, this evidence is consistent with greater liquidity engendering a higher degree of informational efficiency.

An extension of this analysis would be to study the relation between return predictability and illiquidity for fixed income and currency markets. Considering the forecastability of returns from order flows prior to important announcements (when the market might be particularly illiquid due to impending uncertainty) would also be interesting. Pursuit of such topics appears to be a worthwhile agenda for future research.

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